GOVERNMENT OF RAJASTHAN • POLICE HEADQUARTERS, JAIPUR

RAJASTHAN POLICE MARATHON 2027: RUN FOR HER

Official Product Requirement Document (PRD) — Statewide Citizen Engagement, Digital E-Pass, Expo Logistics, & Race Day Command System

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1.0 Executive Summary & Project Vision

Mission Context: The Rajasthan Police Marathon 2027 ('Run for Her') is a statewide mega-initiative organized by the Rajasthan Police Headquarters (PHQ), Jaipur, under the leadership of the Director General of Police (DGP), Rajasthan. The overarching vision of this mission is to foster community-police partnership, promote public health, and champion women's safety, dignity, and empowerment across all 33+ administrative and police districts of Rajasthan.

Key Milestones: The marathon is officially scheduled for Sunday, 03 January 2027, featuring the flagship race starting from Albert Hall (Ram Niwas Garden), continuing through JLN Marg, Jaipur, along with synchronized community runs across all district headquarters statewide. The targeted registration volume exceeds 400,000 participants, encompassing police officers, student delegations, corporate teams, armed constabularies, and general citizens.

2.0 Stakeholder Matrix & User Personas

The system architecture addresses five distinct operational user personas, each with dedicated functional interfaces and security clearance tiers:

Persona Tier	Target User Group	Primary Responsibilities & Portal Actions
Citizen Runner	Public participants, women runners, students, families	Online registration, OTP validation, e-pass generation, WhatsApp sharing, finisher certificate download
Corporate / Bulk Delegate	Institutions, colleges, NCC/NSS cadets, police academies	Bulk CSV registration (up to 500 runners per batch), institutional clearance, nodal officer contact
Expo Counter Volunteer	Designated police personnel & student volunteers at Jaipur Expo	Live pass scanning (Camera/Barcode gun), runner ID verification, RFID kit assignment, anti-fraud kit collection
Gate Marshal	Police security officers stationed at Albert Hall Gantries	Race Day entrance scanning, first-entry authorization, duplicate pass detection, medical triage referral
District SP / PHQ Command	Superintendents of Police (District) & DGP / ADG High Command	Statewide analytics, district CSV export, active runner monitoring, live leaderboard, audit trail inspection

3.0 Functional Requirements Specification (FRS)

3.1 FR-01: Citizen Self-Registration Engine

- **Interface:** Bilingual Registration Form: Primary Hindi interface with English auxiliary labels for comprehensive accessibility.
- **Categories:** Distance Categories: 21 KM Half Marathon (Ages 18+), 10 KM City Run (Ages 16+), and 5 KM Community & Family Run (Open for all age brackets).
- **Runner Data:** Mandatory Citizen Inputs: Full Name, Gender, 10-digit Indian Mobile Number, District, T-shirt Size (XS, S, M, L, XL, XXL), Blood Group, and Emergency Contact Name & Telephone.
- **Health Safety:** Medical Declaration: Runners declare existing health conditions (e.g., cardiac history, asthma, allergies) with voluntary emergency medical notes.

3.2 FR-02: Anti-Fraud Mobile OTP Verification Engine

- **Token Generation:** Automated 6-digit cryptographic OTP generation delivered via SMS Gateway API hook.
- **Security Control:** Sliding-Window Rate Limiting: IP-based and mobile-based rate limiting prevents denial-of-service and SMS bombing attacks (max 3 attempts per minute).
- **Session Token:** Ephemeral Verification Token: A signed JWT token is issued upon successful OTP validation, valid for 15 minutes to finalize registration.

3.3 FR-03: Official Digital E-Pass & Secure Bib Allocation

- **Bib Architecture:** Algorithmic Bib Assignment: Sequential, collision-free chest bib generation with category prefixes (RJ-21K-XXXX, RJ-10K-XXXX, RJ-05K-XXXX).
- **Encrypted QR:** Tamper-Proof QR Code: Encodes HMAC-SHA256 encrypted payload verifying runner identity, category, and certificate token.
- **Graphics Engine:** Client-Side Canvas E-Pass: Pure HTML5 Canvas 2D engine renders official passes with Rajasthan Police emblem, Ashoka Lion Capitol, dynamic text auto-scaling, and instant PNG download.
- **WhatsApp Link:** Direct WhatsApp Dispatch: 1-click WhatsApp API integration sends official pass text and verification links directly to the runner's registered mobile number.

3.4 FR-04: Expo Counter Logistics & RFID Kit Handover

- **Scanner Engine:** Multi-Device Camera Scanner: WebRTC engine with platform-aware facing mode (mobile back camera vs PC webcam) and 400ms driver deallocation delay.
- **Barcode Gun:** Hardware Barcode Gun Support: Native input focus captures USB/Bluetooth 2D barcode scanner keystrokes in under 100 milliseconds.
- **Anti-Fraud:** Single-Handover Locking: Database atomicity prevents duplicate kit distribution. If a runner's bib was already collected, the terminal sounds an audible fraud alert displaying prior collection timestamp and officer badge ID.

3.5 FR-05: Albert Hall Race Day Gate Security & Gantry Access

- **Access Control:** Albert Hall Entry Verification: Gate marshals scan e-passes at stadium gantries. System logs first entry and instantly flags duplicate gate entries.
- **Live Logging:** Real-Time Gate Logs: Audit table tracks gate entry timestamps, operator IDs, and sector densities.

3.6 FR-06: Finisher Certificate & Timing Mat Hub

- **Search:** Instant Certificate Lookup: Public retrieval by Bib number, Mobile number, or Certificate ID.
- **Verification:** Tamper-Proof Verification: Cryptographic certificate ID verifies authenticity against the central database, displaying official rank, net finish time, and DGP digital signature.
- **Print Engine:** Isolated A4 Landscape Print Engine: Dedicated print stylesheets strip away all website UI, headers, and footers, printing only the A4 landscape certificate.

3.7 FR-07: High Command Admin Portal & District SP Desks

- **RBAC:** Role-Based Access Control (RBAC): PHQ State Command has statewide jurisdiction; District SPs access their district data; Expo volunteers access counter desks.
- **CSV Engine:** Database CSV Export: Instant export of sanitized runner data for law enforcement record-keeping and timing mat synchronization.

4.0 Non-Functional Requirements (NFR)

NFR Domain	Technical Benchmark	Implementation Architecture
System Scalability	400,000+ registrations, 15,000 req/min peak	Turso Cloud Distributed Edge SQLite + Vercel Serverless auto-scaling
Page Load Speed	First Contentful Paint < 1.0s, TTI < 1.8s	Vite 8 Rollup manual vendor chunking, compressed assets, tree-shaking
Security & Privacy	OWASP Top 10 compliance, A+ SSL Rating	W3C Permissions-Policy camera=(self), parameterized SQL queries, zero plaintext secrets
Cross-Platform	Windows, Mac, iOS Safari, Android Chrome	Responsive CSS flexbox/grid layout, WebRTC multi-device fallback, 2D input wedge
Reliability & DR	99.95% uptime during event week	Dual-mode cloud edge + offline local storage failover for volunteer terminals
Accessibility	WCAG 2.1 Level AA Compliance	Bilingual Hindi/English typography, high-contrast palette, screen-reader aria attributes

5.0 Verification & Acceptance Criteria Matrix

Feature Module	Acceptance Criteria	Verification Result
Citizen Registration	Form submits, validates inputs, generates valid bib without race conditions	VERIFIED (100% Pass)
OTP Engine	Generates 6-digit code, enforces 45s timer, rejects expired/invalid codes	VERIFIED (100% Pass)
E-Pass Rendering	Bib number scales dynamically within 280px frame without text clipping	VERIFIED (100% Pass)
Expo Scanner	Webcam/Mobile camera streams live video, captures QR code, populates runner card	VERIFIED (100% Pass)
Double-Kit Protection	Second attempt to distribute kit for same bib returns ALREADY_COLLECTED error	VERIFIED (100% Pass)
Certificate Print	Ctrl+P prints clean A4 landscape certificate with zero web UI clutter	VERIFIED (100% Pass)
Admin Portal	District filtering isolates district participants; CSV export generates clean file	VERIFIED (100% Pass)

6.0 Sign-off & Approval

Formal Approval: This Product Requirement Document (PRD) represents the official baseline for the Rajasthan Police Marathon 2027 technical implementation. All requirements specified herein have been verified, validated, and approved for production operations.